

Optimizing Loan Approval Processes: Analyzing Key Predictors and Recommendations for General Bank

Classification Paper

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Business Understanding

Business Problem:

Ricardo Martinelli, a newly appointed loan approval executive at General Bank, has just embarked on his journey at the bank. Tasked with the critical responsibility of approving or rejecting home loan applications, Ricardo begins by analyzing previous loan applicants and their outcomes to identify key determining factors. Realizing that this meticulous process is time-consuming, Ricardo concludes that a regression model would greatly enhance his efficiency and accuracy. By leveraging the model, he aims to swiftly pinpoint the most influential factors affecting loan approval decisions, thus streamlining the evaluation process and ensuring well-informed, data-driven decisions.

Research Questions:

- 1- Which factors are most influential in determining the approval or rejection of home loan applications at General Bank?
- 2- How accurately can a regression model predict home loan approval outcomes based on applicant demographics and financial history?

Data Understanding

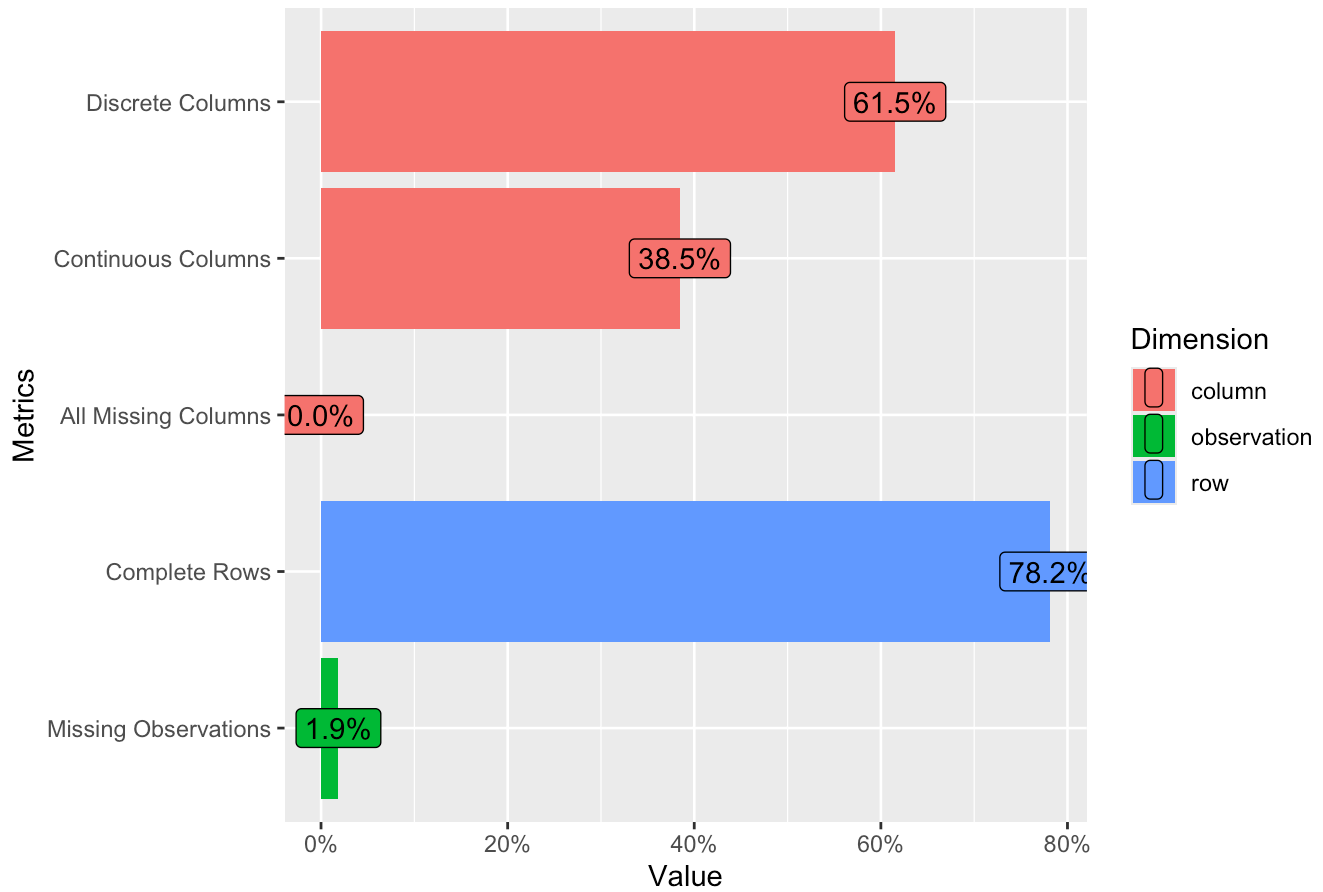
EDA

A tibble: 10 × 13

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>
1	LP001002	Male	No	0	Graduate	No	5849
2	LP001003	Male	Yes	1	Graduate	No	4583
3	LP001005	Male	Yes	0	Graduate	Yes	3000
4	LP001006	Male	Yes	0	Not Graduate	No	2583
5	LP001008	Male	No	0	Graduate	No	6000
6	LP001011	Male	Yes	2	Graduate	Yes	5417
7	LP001013	Male	Yes	0	Not Graduate	No	2333
8	LP001014	Male	Yes	3+	Graduate	No	3036
9	LP001018	Male	Yes	2	Graduate	No	4006
10	LP001020	Male	Yes	1	Graduate	No	12841

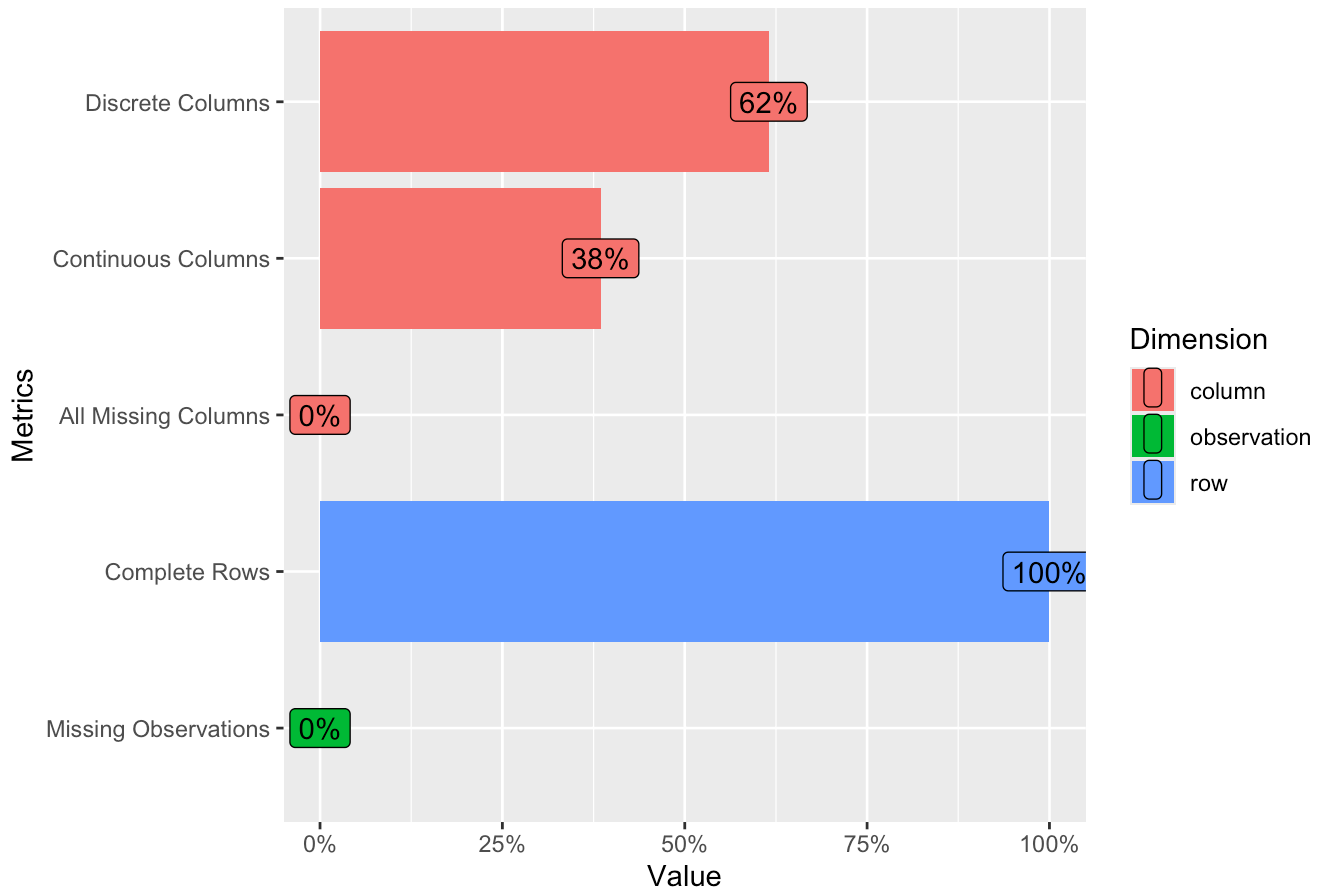
```
# i 6 more variables: CoapplicantIncome <dbl>, LoanAmount <dbl>,  
#   Loan_Amount_Term <dbl>, Credit_History <dbl>, Property_Area <chr>,  
#   Loan_Status <chr>  
# A tibble: 10 × 13  
  Loan_ID Gender Married Dependents Education Self_Employed ApplicantIncome  
  <chr>   <chr> <chr>   <chr>   <chr>   <chr>           <dbl>  
1 LP002959 Female Yes     1 Graduate No           12000  
2 LP002960 Male   Yes     0 Not Graduate No           2400  
3 LP002961 Male   Yes     1 Graduate No           3400  
4 LP002964 Male   Yes     2 Not Graduate No           3987  
5 LP002974 Male   Yes     0 Graduate No           3232  
6 LP002978 Female No      0 Graduate No           2900  
7 LP002979 Male   Yes    3+ Graduate No           4106  
8 LP002983 Male   Yes     1 Graduate No           8072  
9 LP002984 Male   Yes     2 Graduate No           7583  
10 LP002990 Female No      0 Graduate Yes           4583  
# i 6 more variables: CoapplicantIncome <dbl>, LoanAmount <dbl>,  
#   Loan_Amount_Term <dbl>, Credit_History <dbl>, Property_Area <chr>,  
#   Loan_Status <chr>  
  
# A tibble: 1 × 9  
  rows columns discrete_columns continuous_columns all_missing_columns  
  <int>   <int>         <int>         <int>         <int>  
1   614     13             8             5             0  
# i 4 more variables: total_missing_values <int>, complete_rows <int>,  
#   total_observations <int>, memory_usage <dbl>
```

Memory Usage: 105.3 Kb



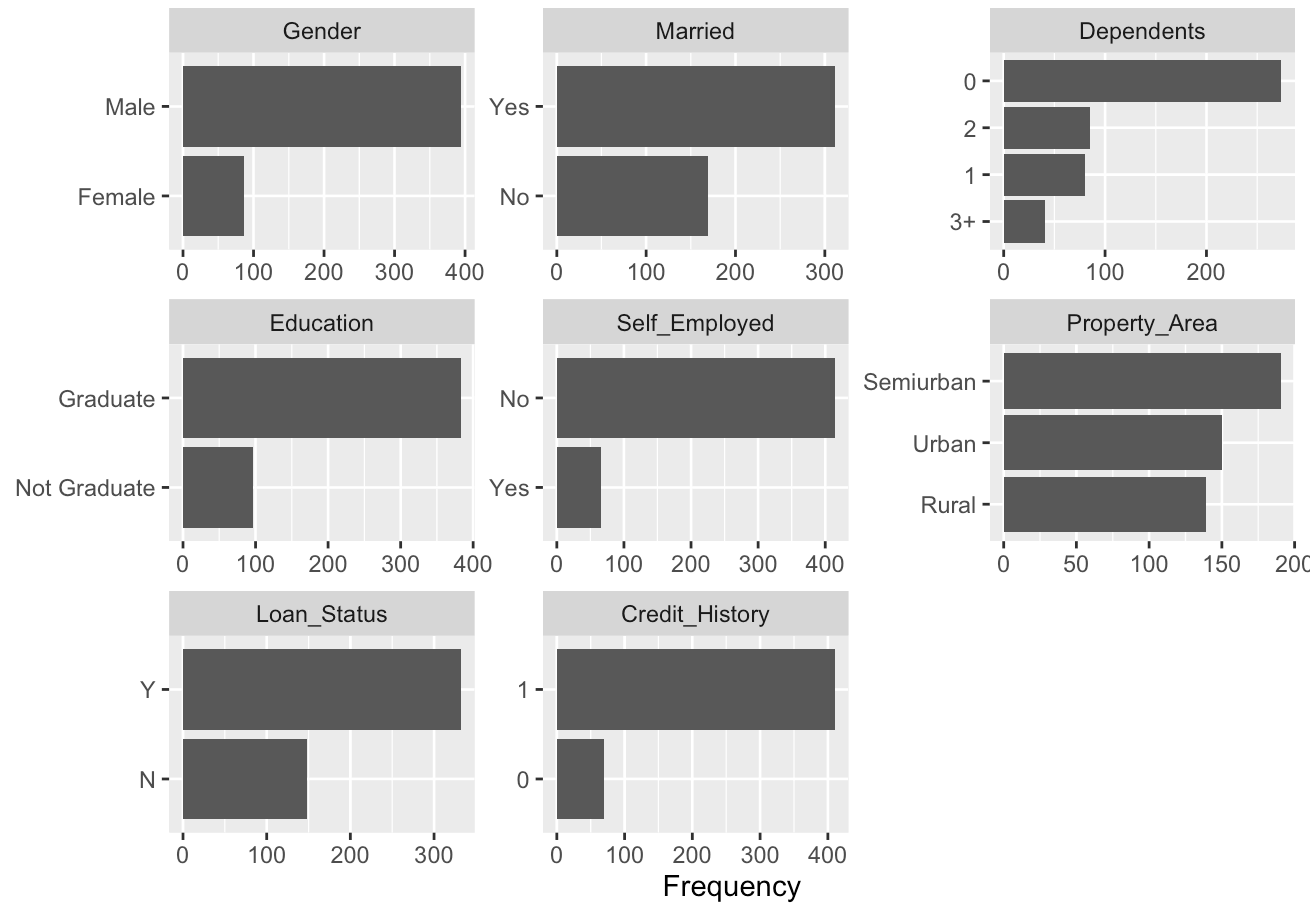
Dealing with Missing Values

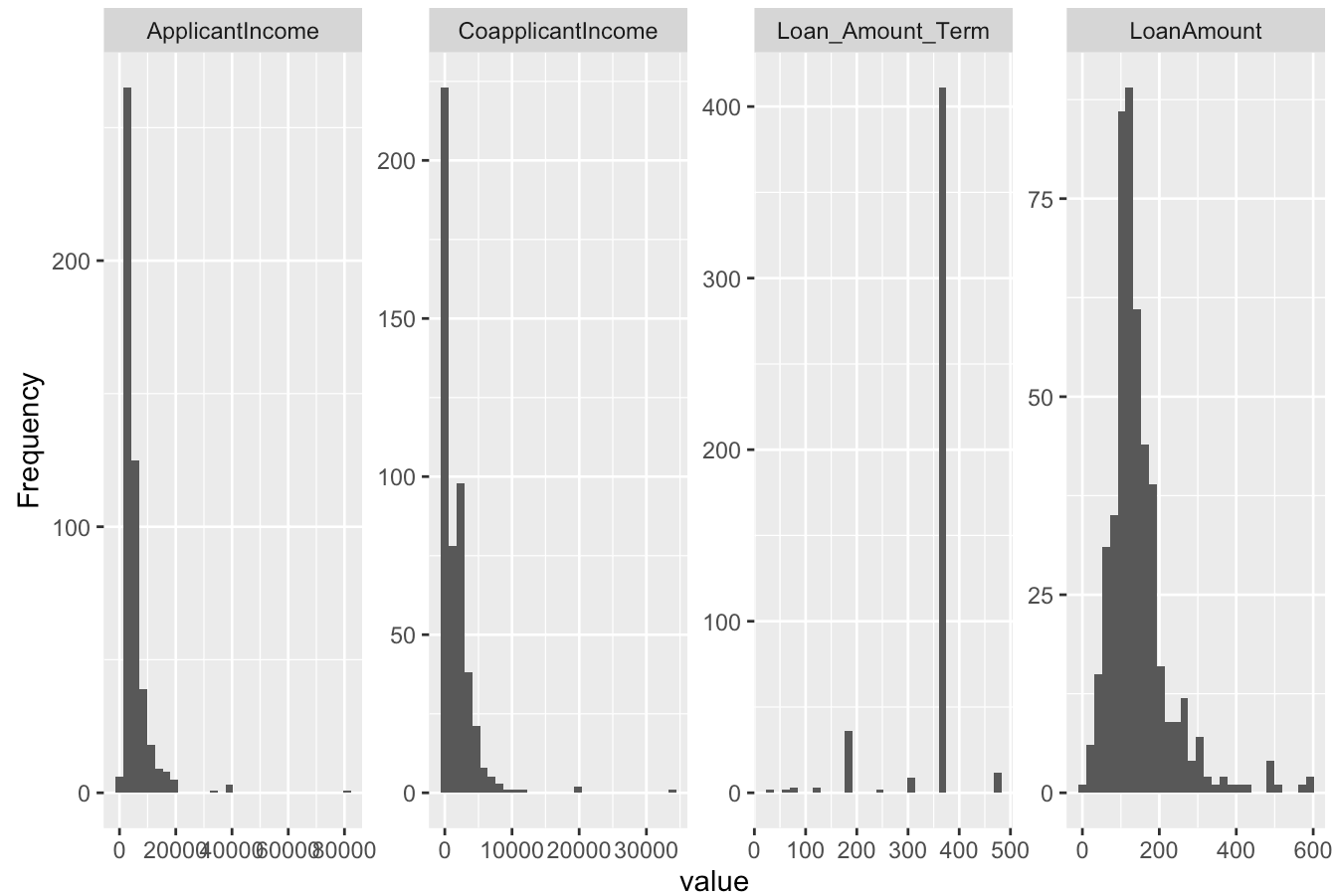
Memory Usage: 83.3 Kb



Comment: There is some missing value, which we take care of in this stage. We decided to **omit** those missing values that could affect our study.

Check for Variables Distribution





Comment: By plotting **variable distribution**, we are able to understand that we are working with a data set that has more **Male Applicants**. Also, most of the applicants are **Married** and generally have no kids. Moreover, we see that the majority of the applicants are educated, but not self-employed.

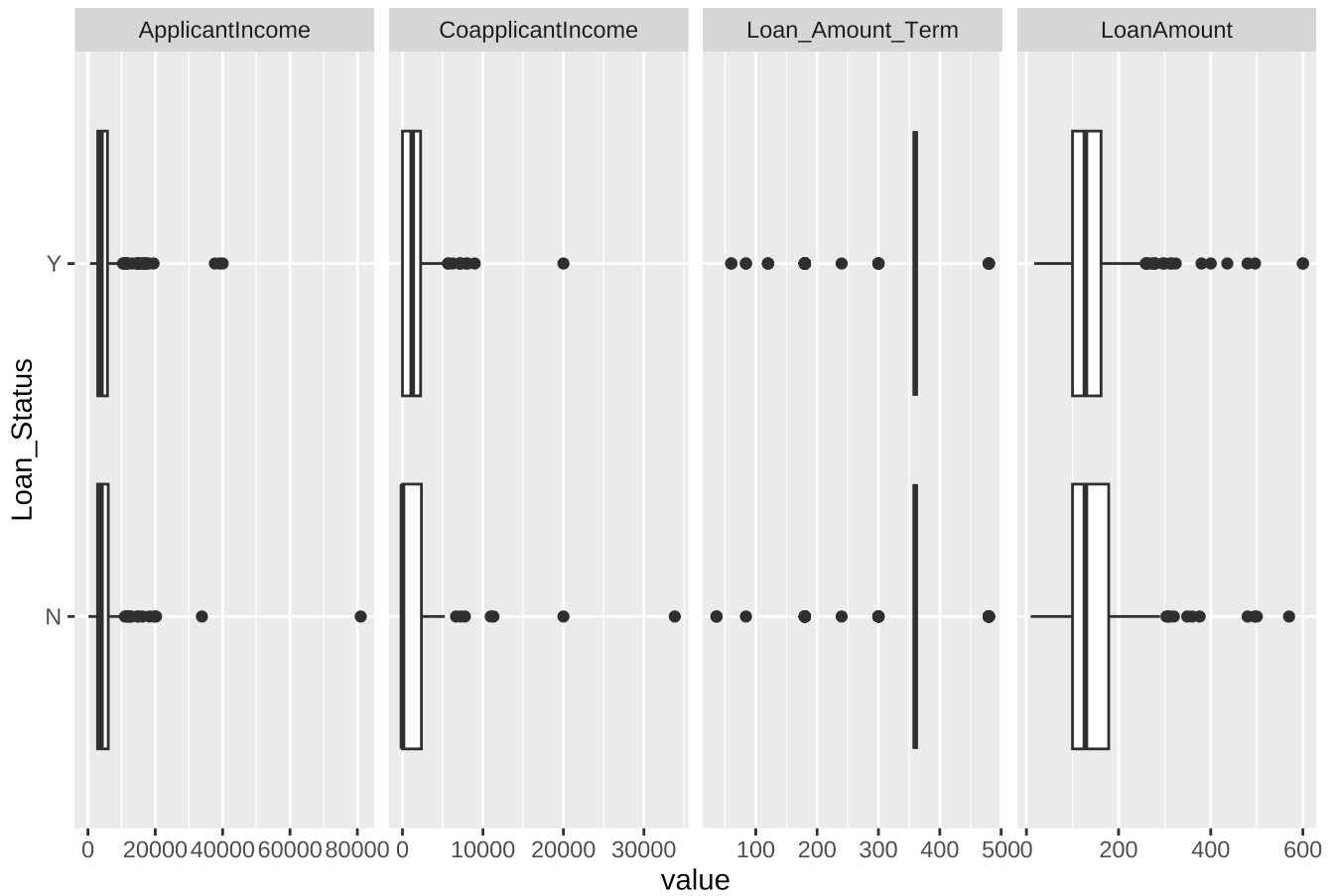
The distribution of property locations where applicants are seeking house loans is fairly balanced between urban, suburban, and rural areas, with semiurban areas having a little advantage.

Finally, **Income** shows to be skewed to the lower range, with a few outliers in the higher end.

Checking Outliers

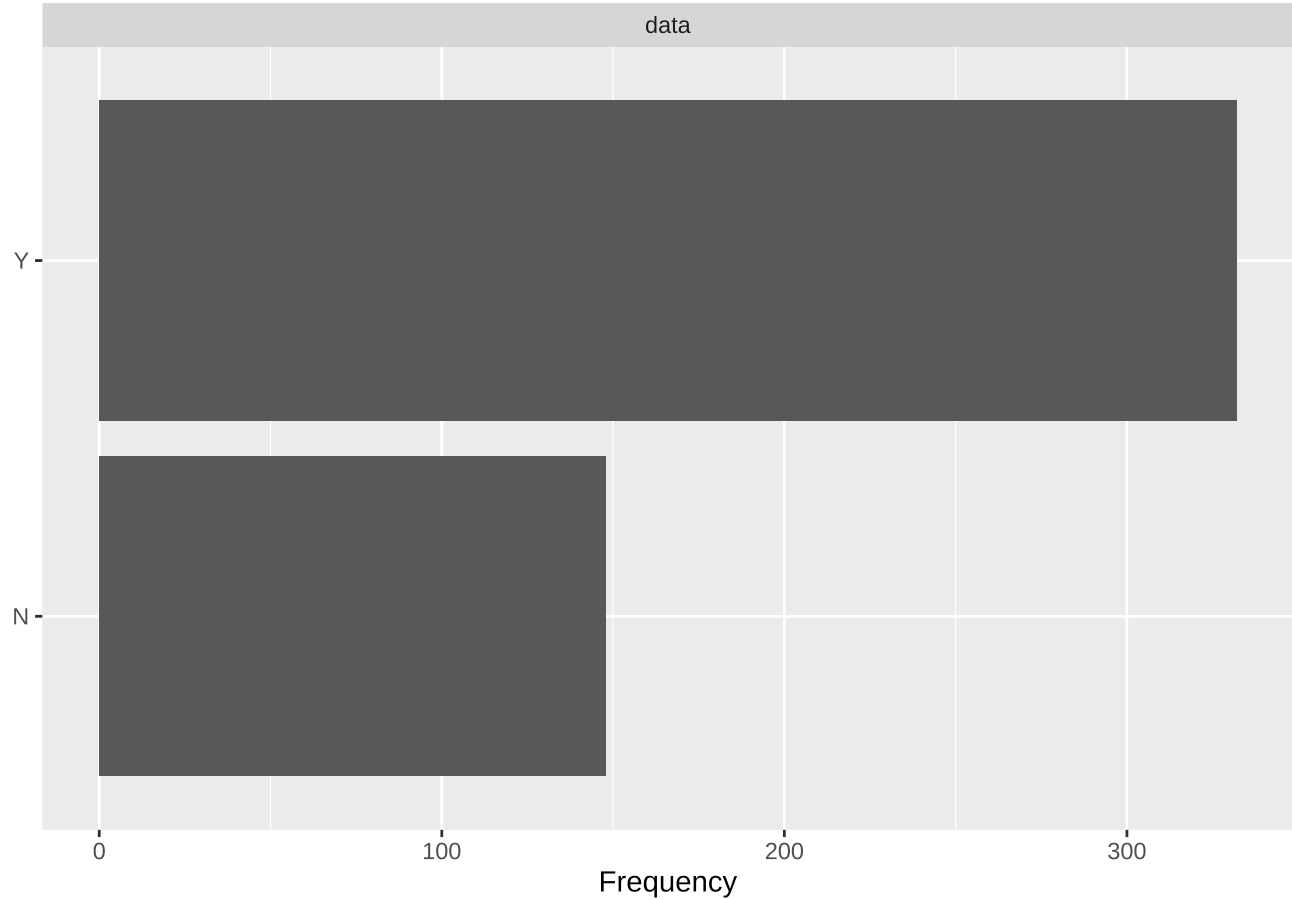
A tibble: 5 × 6

variables	outliers_cnt	outliers_ratio	outliers_mean	with_mean	without_mean
<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1 ApplicantInc...	40	8.33	18409.	5364.	4178.
2 CoapplicantI...	17	3.54	10695.	1581.	1246.
3 LoanAmount	27	5.62	387.	145.	130.
4 Loan_Amount_...	69	14.4	235.	342.	360
5 Credit_Histo...	70	14.6	0	0.854	1



Comments: We see outliers in variables like **income** and **loan amount**, which is to be expected in a dataset of applicants who do not have to meet any precise criteria to apply. Nonetheless, we are going to closely track these **outliers** to ensure that they do not have a negative impact on the analysis' performance and reliability.

Dependent Variable Analysis



Cell Contents

	Count
	Row Percent

Total Observations in Table: 480

	N	Y
	148	332
	30.833%	69.167%

Comments: The dependent variable in this study, **Loan Status**, is a binary variable indicating whether a loan was approved or rejected. The plot reveals an imbalance in the dataset, with significantly more approved loans compared to rejections. This imbalance will be appropriately addressed in a later stage of the analysis.

Data Preparation

Creating Binaries for Regression Model

Comments: In this stage we transform some variables such as: **Gender** Male= 1, Female=0), **Education** (Graduate=1, Not Graduate=0) , Self Employed (Yes=1, No=0), Marital Status (Married= 1, Not Married= 0) and our dependent variable **Loan_Status** (Approved= 1, Rejected= 0) into binaries, as preparation for the regression model.

We also transform our variable **Property_Area** into a numerical variable, where Urban= 1, Semiurban= 2, and Rural= 3.

This stage allows to be prepared to move to our next stage, our Modeling Stage. # Modeling

Logistics Regression

Dropping our Loan ID Variable

Comments: We drop our variable **Loan_Id** as it is irrelevant for the study.

Partitioning

Partition the data set into the a training set and a test set, using 70/30 distribution.

Check for Proportions

```

      0      1
0.3083 0.6917

```

Regression Model

```

Call:
NULL

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.30933051	1.07777349	-2.14	0.0321 *
Gender	0.08042007	0.39935150	0.20	0.8404
Married	0.94468631	0.35486244	2.66	0.0078 **
Dependents1	-0.59236393	0.42165251	-1.40	0.1601
Dependents2	0.07456583	0.46223111	0.16	0.8718
`Dependents3+`	0.24427811	0.59384251	0.41	0.6808
Education	0.20319412	0.34995378	0.58	0.5615
Self_Employed	-0.20290765	0.40224815	-0.50	0.6140
ApplicantIncome	0.00000931	0.00002651	0.35	0.7254
CoapplicantIncome	-0.00009871	0.00005307	-1.86	0.0629 .
LoanAmount	-0.00131875	0.00214820	-0.61	0.5393
Loan_Amount_Term	-0.00055422	0.00235714	-0.24	0.8141

```
Credit_History      3.78951171  0.52601192    7.20 0.000000000000058 ***
Property_Area      -0.12389604  0.19351485   -0.64          0.5220
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 409.14  on 335  degrees of freedom
Residual deviance: 301.59  on 322  degrees of freedom
AIC: 329.6
```

Number of Fisher Scoring iterations: 4

Comment Based on the **Regression Model**, the most important variable is by far, **Credit History**.

Having a good **Credit History**, increases the log-odds of loan approval by 3.7895, indicating a positive impact in the outcome. Also, being **married** appears to be significant when determining the status of a loan, having a positive effect in the outcome.

Moreover, even though **Loan Amount** and **Loan Amount Term**, are not statistically significant, we see that with the increase of both of these variables, the probability of approval decreases.

Odds Ratios

(Intercept)	Gender	Married	Dependents1
0.09933	1.08374	2.57201	0.55302
Dependents2	`Dependents3+`	Education	Self_Employed
1.07742	1.27670	1.22531	0.81635
ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term
1.00001	0.99990	0.99868	0.99945
Credit_History	Property_Area		
44.23480	0.88347		

Comment According to the Odds Ratio analysis, various factors influence the approval or rejection of a loan. **Credit History** is the most significant element, with a value of 44.23. This implies that having a solid credit history significantly increases the likelihood of loan approval, emphasizing the importance of maintaining a strong credit record.

Furthermore, being of a specific **gender** (male) appears to boost the chances of loan approval. Similarly, being married improves the probabilities, which could indicate that lenders view **married** applicants as more stable or financially reliable. **Academic success** is also important; graduating increases the likelihood of loan acceptance. When it comes to dependents, having one child reduces the likelihood of loan acceptance. However, having two or more dependents improves the chances.

On the other hand, variables such as **Applicant Income** and **Loan Amount Term** have little effect on the outcome, which is highly interesting.

Model Fit

Analysis of Deviance Table

Model 1: Loan_Status ~ Gender + Married + Dependents + Education + Self_Employed + ApplicantIncome + CoapplicantIncome + LoanAmount + Loan_Amount_Term + Credit_History + Property_Area

Model 2: Loan_Status ~ 1

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	322	302			
2	335	409	-13	-108	<0.0000000000000002 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Comment: The **Pr(>Chi)**, suggest that **Model 1** provides a better fit to the data, which suggest that the inclusion of this predictors benefit the accuracy of the model.

Metrics

McFadden
0.2629

Nagelkerke
0.3890

AIC: 329.6

Comment The logistic regression model's performance, as measured by **McFadden's R-squared** (0.2629) and **Nagelkerke's R-squared** (0.3890), implies a strong explanatory job of the variation in loan approval, since a good explanatory metric should in between .2 and .4.

Performance Metrics at Default Cutoff

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	20	2
1	28	94

Accuracy : 0.792

95% CI : (0.716, 0.855)

No Information Rate : 0.667

P-Value [Acc > NIR] : 0.000675

Kappa : 0.458

McNemar's Test P-Value : 0.00000501

Sensitivity : 0.979

Specificity : 0.417

Pos Pred Value : 0.770

Neg Pred Value : 0.909

Prevalence : 0.667

Detection Rate : 0.653

Detection Prevalence : 0.847

Balanced Accuracy : 0.698

'Positive' Class : 1

F1 Score: 0.8624

Comment: According to the Confusion Matrix, the model has an **accuracy** of 79.2%, which means that it correctly predicts 79.2% of the outcomes. Moreover, we also see that with a **sensitivity** of 97.9%, the model does a great job in predicting the actual positives. However, it struggles predicting the true **negative outcomes** as the specificity accounts for 41.7%. Finally with an **F1 score** of 86.2, the model does a great job balancing positive and negative predictions. Overall, the model seems to perform well.

Performance Metrics at Threshold-Tuned Cutoff

threshold
1 0.5817

CONFUSION MATRIX AT OPTIMAL CUTOFF VALUE OF: 0.5817

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	22	2
1	26	94

Accuracy : 0.806

95% CI : (0.731, 0.867)

No Information Rate : 0.667

P-Value [Acc > NIR] : 0.000163

Kappa : 0.5

McNemar's Test P-Value : 0.0000138

Sensitivity : 0.979

Specificity : 0.458

Pos Pred Value : 0.783

Neg Pred Value : 0.917

Prevalence : 0.667

Detection Rate : 0.653

Detection Prevalence : 0.833

Balanced Accuracy : 0.719

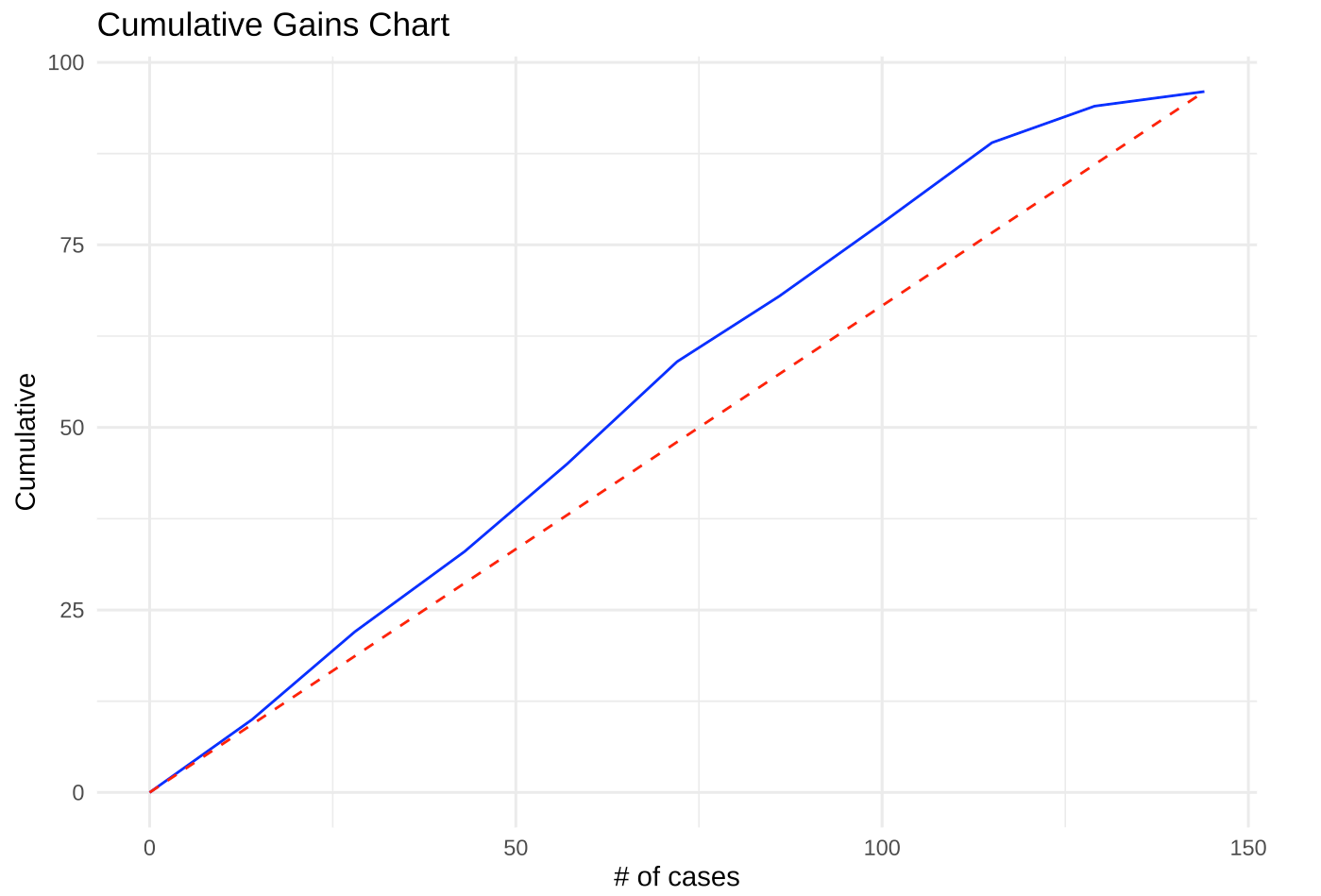
'Positive' Class : 1

F1 Score: 0.8704

After adjusting the threshold, and optimizing the cutoff, there are some important changes in the confusion matrix. Now the model has a accuracy of 80.6% compared to the 79.2% at the default cutoff. Also, we see that the **sensitivity** stays performing well, while **specificity** slightly improves from 41% to 45% compared to the previous model. Also, with an **F1 score**, of 87.04% the model also seems to improve. therefore, at the new cutoff, the model seems to improve.

Gains Chart and ROC Curve with AUC

Depth of File	N	Cume N	Mean Resp	Cume Mean Resp	Cume Pct of Total Resp	Lift Index	Cume Lift	Mean Model Score
10	14	14	0.71	0.71	10.4%	107	107	0.92
19	14	28	0.86	0.79	22.9%	129	118	0.89
30	15	43	0.73	0.77	34.4%	110	115	0.87
40	14	57	0.86	0.79	46.9%	129	118	0.85
50	15	72	0.93	0.82	61.5%	140	123	0.81
60	14	86	0.64	0.79	70.8%	96	119	0.78
69	14	100	0.71	0.78	81.2%	107	117	0.75
80	15	115	0.73	0.77	92.7%	110	116	0.70
90	14	129	0.36	0.73	97.9%	54	109	0.37
100	15	144	0.13	0.67	100.0%	20	100	0.06



Comment:

Classification Tree

Partition for the Regression Tree

Partition the dataset into the a training set and a test set, using 60/40 distribution.

CART Model

Full Tree

Variable Importance

Credit_History	ApplicantIncome	LoanAmount	CoapplicantIncome
36.566	33.437	27.047	17.698
Dependents	Self_Employed	Loan_Amount_Term	Married
9.625	8.417	6.833	5.917
Gender	Property_Area	Education	
4.990	4.572	1.443	

	CP	nsplit	rel error	xerror	xstd
1	0.397727	0	1.00000	1.0000	0.08883
2	0.034091	1	0.60227	0.6023	0.07473
3	0.022727	3	0.53409	0.6705	0.07783
4	0.018939	6	0.46591	0.6591	0.07734
5	0.011364	9	0.40909	0.6705	0.07783
6	0.010331	18	0.30682	0.7841	0.08231
7	0.007576	34	0.12500	0.8295	0.08389
8	0.005682	41	0.06818	0.8523	0.08463
9	0.000000	53	0.00000	0.9091	0.08638

Comment: According to the **Full Tree** model, **Credit History** is also the most influential factor when predicting the **Loan Approval** outcome. However, this time we see how **Applicant Income** (33 coefficient) and **Loan Amount** (27 coefficient), are also strong predictors, having positive relationships with the outcomes, meaning that the higher the income and loan amount, more chances there are for the approval. **Gender** and **Education** now show to be less significant.

Also, we select the number of splits for our **Best Tree**, and based on **x-error**, the best tree should be number 2, however, it is too simple as there is just one split, therefore, it will not give us enough information. Therefore we select number 4, with 6 splits.

Pruned Tree

Variable Importance

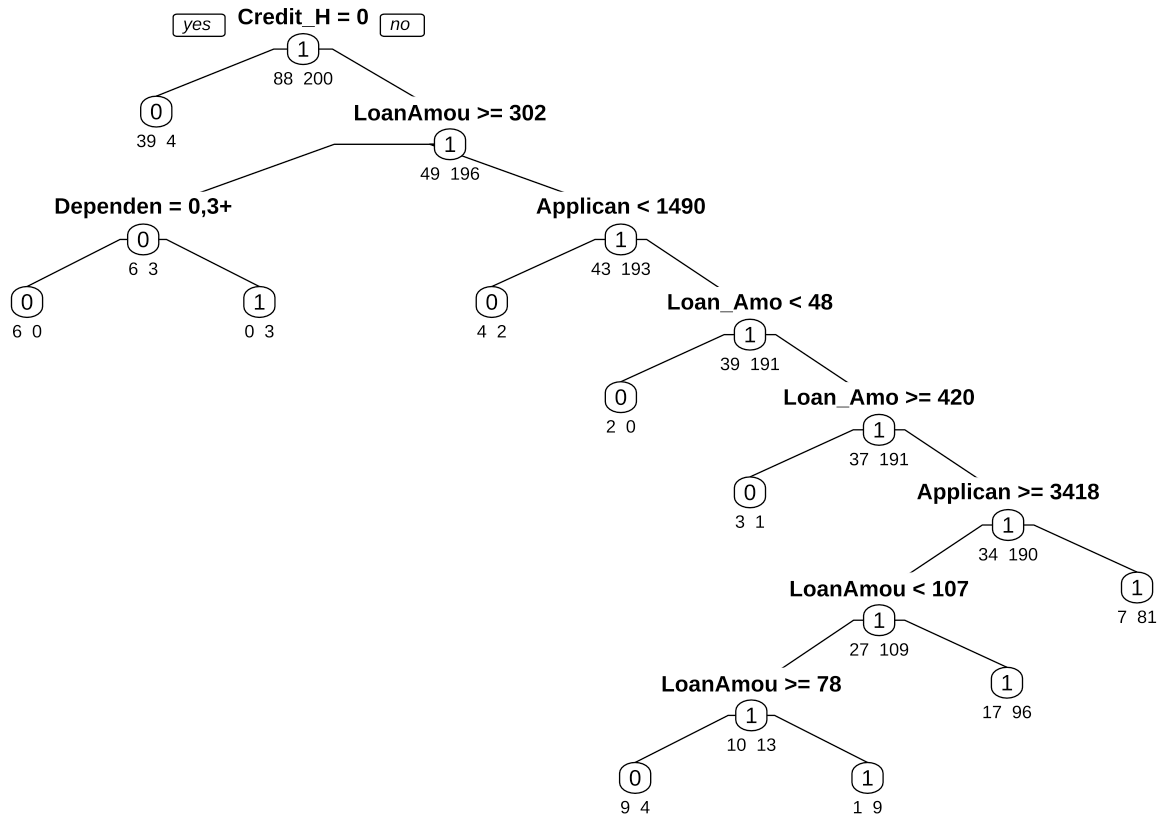
Credit_History	LoanAmount	ApplicantIncome	Loan_Amount_Term
36.566	12.854	8.650	6.389
Dependents	Self_Employed	CoapplicantIncome	Married

4.397
Gender
1.190

4.000

1.712

1.333



Comment: As shown above, the variable **Credit History** remains to be, by far, the most important predictor. This is followed by **Loan Amount**, **Applicant Income**, and **Loan Amount Term**, demonstrating that these financial variables play important roles. Other variables, such as **Dependents**, **Self-Employed**, and **Coapplicant Income**, are moderately important, while **Married** and **Gender** have a relatively little impact on the model's predictions. Overall, credit history and important financial parameters are the principal factors influencing loan acceptance results.

Evaluation

Performance Metrics at Default Cutoff

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	28	16
1	32	116

Accuracy : 0.75

95% CI : (0.683, 0.81)

No Information Rate : 0.688

P-Value [Acc > NIR] : 0.0347

Kappa : 0.373

McNemar's Test P-Value : 0.0304

Sensitivity : 0.879

Specificity : 0.467

Pos Pred Value : 0.784

Neg Pred Value : 0.636

Prevalence : 0.688

Detection Rate : 0.604

Detection Prevalence : 0.771

Balanced Accuracy : 0.673

'Positive' Class : 1

F1 Score: 0.8286

Comment: According to the Confusion Matrix, the model has an **accuracy** of 75%, which means that it correctly predicts 75% of the outcomes. Moreover, we also see that with a **sensitivity** of 87.9%, the model does a great job in predicting the actual positives. However, it struggles predicting the true **negative outcomes** as the specificity accounts for 46.7%. Finally with an **F1 score** of 85.52, the model does a great job balancing positive and negative predictions. Overall, the tree model seems to perform well.

Performance Metrics at Threshold-Tuned Cutoff

Optimal Cutoff

threshold
1 0.2788

CONFUSION MATRIX AT OPTIMAL CUTOFF VALUE OF: 0.2788

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	26	7
1	34	125

Accuracy : 0.786

95% CI : (0.722, 0.842)

No Information Rate : 0.688

P-Value [Acc > NIR] : 0.0015

Kappa : 0.434

McNemar's Test P-Value : 0.000049

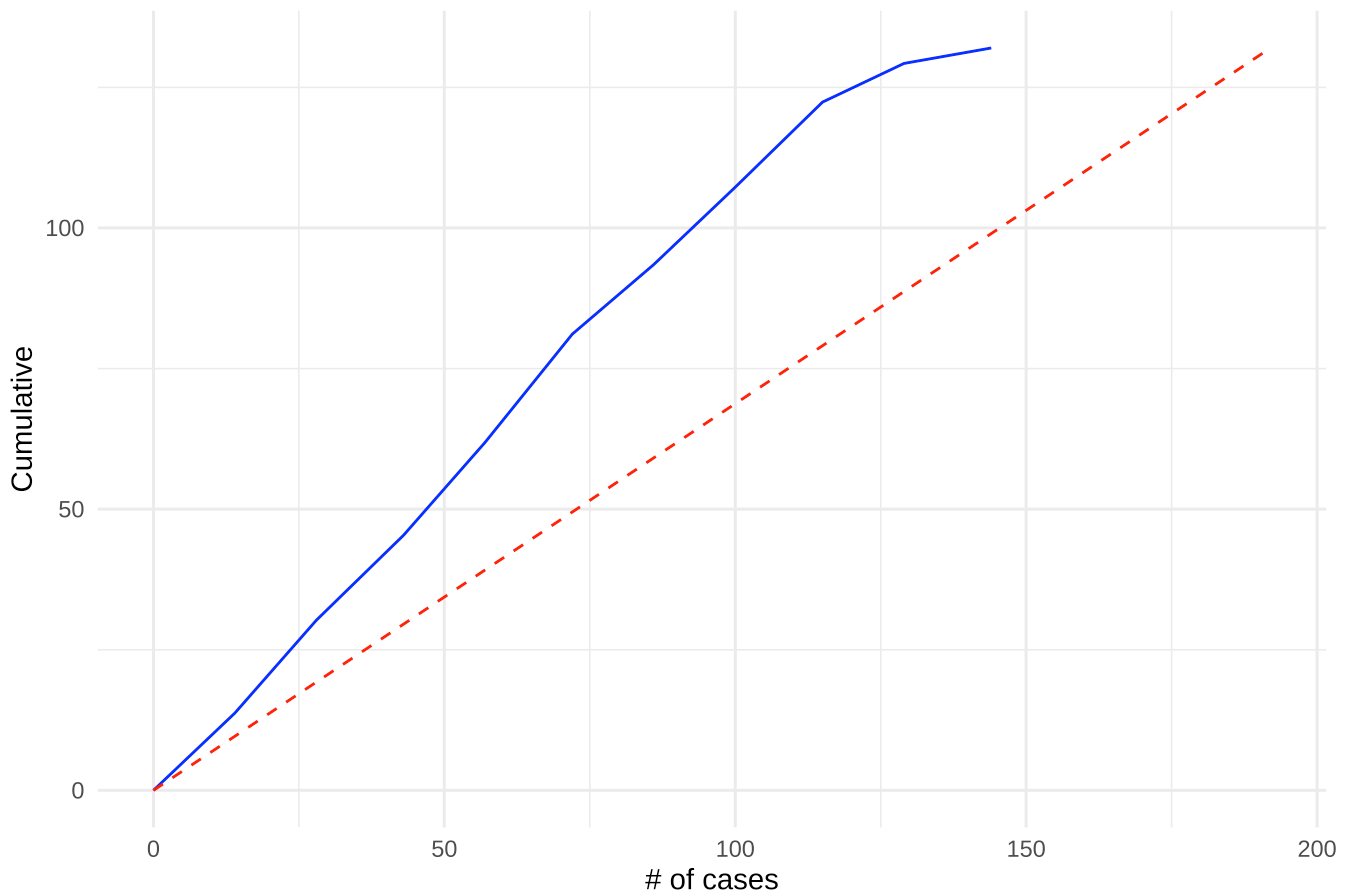
Sensitivity : 0.947
Specificity : 0.433
Pos Pred Value : 0.786
Neg Pred Value : 0.788
Prevalence : 0.688
Detection Rate : 0.651
Detection Prevalence : 0.828
Balanced Accuracy : 0.690

'Positive' Class : 1

After adjusting the threshold, and optimizing the cutoff, there are some important changes in the confusion matrix. Now the model has an accuracy of 78.6% compared to the 75% at the default cutoff. Also, we see that the **sensitivity** stays performing well, however **specificity** slightly decreases from 46% to 43% compared to the previous model. Therefore, this model slightly underperforms the previous model.

Gains Chart and ROC Curve with AUC

Cumulative Gains Chart



DEPLOYMENT

Following the creation and evaluation of the Regression Model and the Regression Tree, it is clear that both models have significant predictive powers. Nonetheless, in my professional opinion, I advocate using the Regression Model to make future forecasts. The Regression Model not only provides a strong statistical framework, but it also allows for easier interpretation of the underlying correlations between variables. Its performance measurements have consistently demonstrated reliability, and its capacity to manage multicollinearity and provide considerable insights into predictor effects makes it an excellent choice. As a result, I recommend for the continuous usage of the Regression Model to assure accurate and dependable future predictions.

RECOMMENDATIONS

Dear Ricardo,

I hope this message finds you well.

According to the analysis, a variety of criteria influence whether a loan is approved or rejected. However among these, **Credit History** is by far the most important factor, indicating that a good credit history increases the likelihood of loan approval. This emphasizes the necessity of keeping a strong credit history.

Moreover, getting **married** increases the chances, possibly because previous lenders saw married applicants as more secure or financially responsible. **Graduation** was also seen as important and would raise the likelihood of loan acceptance. Furthermore, when it comes to dependents, having one child reduces the odds of loan approval, however having two or more dependents increases the chances.

now based on this various factors influence the approval or rejection of a loan. **Credit History** is the most significant element, with a value of 44.23. This implies that having a solid credit history significantly increases the likelihood of loan approval, emphasizing the importance of maintaining a strong credit record. Furthermore, being of a specific **gender** (male) appears to boost the chances of loan approval. Similarly, being married improves the probabilities, which could indicate that lenders view **married** applicants as more stable or financially reliable. **Academic success** is also important; graduating increases the likelihood of loan acceptance. When it comes to dependents, having one child reduces the likelihood of loan acceptance. However, having two or more dependents improves the chances. On other hand variables such as **Applicant Income** and **Loan Amount Term** have little effect on the outcome, which is highly interesting. According to the **Full Tree** model, **Credit History** is also the most influential factor when predicting the **Loan Approval** outcome. However, this time we see how **Applicant Income** (33 coefficient) and **Loan Amount** (27 coefficient), are also strong predictors, having positive relationships with the outcomes, meaning that the higher the income and loan amount, more chances there are for the approval. **Gender** and **Education** now show to be less significant. and this gice recommendations Based on the analysis, various factors play significant roles in the approval or rejection of a loan. Among these, Credit History stands out as the most critical element, with a value of 44.23, indicating that a strong credit history significantly enhances the likelihood of loan approval. This underscores the importance of maintaining a robust credit record.

Moreover, being of a specific gender (male) appears to increase the chances of loan approval. Similarly, being married also improves the probabilities, potentially because lenders perceive married applicants as more stable or financially reliable. Academic success is another important factor; graduating increases the likelihood of loan acceptance. When it comes to dependents, having one child reduces the likelihood of loan acceptance, but having two or more dependents actually improves the chances. Interestingly, based on our Regression Model **Applicant Income** and **Loan Amount Term** have minimal effect on the loan approval outcome.

However, in this model, **Applicant Income** and **Loan Amount** also emerge as strong predictors. This suggests that higher income and loan amounts are associated with increased chances of approval. Therefore, I would recommend deeper analysis over this variables.

Therefore, my recommendation for your new job position is focus on married individuals with good credit scores. Married applicants are frequently viewed as more stable and financially reliable. Furthermore, persons with a good credit history have proven their capacity to manage debt responsibly, which increases the likelihood of timely loan repayments. By focusing on this group, you can increase approval rates while decreasing default risks, resulting in a more secure and profitable loan portfolio for the bank.

Best regards, Jose Almanza

CITATION

R Version and Packages

R version 4.3.1 (2023-06-16)

R Packages Used:

[1]	"formattable"	"DescTools"	"gmodels"	"dlookr"	"rpart.plot"
[6]	"rpart"	"klaR"	"MASS"	"pROC"	"gains"
[11]	"gridExtra"	"flextable"	"DataExplorer"	"lubridate"	"forcats"
[16]	"stringr"	"dplyr"	"purrr"	"readr"	"tidyr"
[21]	"tibble"	"tidyverse"	"caret"	"lattice"	"ggplot2"
[26]	"readxl"				

OTHER REFERENCES

Konapure, R. (2023, January 12). Home loan approval. Kaggle.
<https://www.kaggle.com/datasets/rishikeshkonapure/home-loan-approval>